

***Theme > recipient* double object constructions and hierarchy effects in English**

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1. Introduction

Varieties of English spoken in Northern England have Double Object Constructions (DOCs) with objects in both the standard order (1a)—the recipient precedes the theme—and inverse order (1b)—the theme precedes the recipient. The readings of both examples are equivalent to the prepositional dative in (1c).

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|-----|----|-----------------------|-----------------------------|
| (1) | a. | They showed me it. | <i>recipient > theme</i> |
| | b. | They showed it me. | <i>theme > recipient</i> |
| | c. | They showed it to me. | <i>prepositional dative</i> |

Formal accounts of this DOC order focus on the broader issue of symmetry (Haddican 2010, Haddican and Holmberg 2012, 2019, Holmberg et al. 2019). The goal of this paper is to provide an Agree-based account of both DOC orders, their object size constraints (Section 3) and their hierarchy effects (Section 4).

I propose that the observed data results from the reordering of weak pronouns during the process of instantiating hierarchy effects. In *recipient > theme* orders the recipient moves to a higher position to satisfy an [EPP] feature on the probing head and the theme moves to avoid stress. In *theme > recipient* orders the recipient remains in its base position due to the lack of an [EPP] feature and the theme moves to avoid stress, raising over the recipient and establishing the inverted object order.

2. DOC Status

Verbs differ with regard to their acceptability in *recipient > theme* DOCs and prepositional dative constructions. *Theme > recipient* DOCs align with *recipient > theme* DOCs rather than with prepositional datives (Haddican 2010). Verbs like ‘donate’ are unacceptable with DOCs (2), while verbs like ‘deny’ are unacceptable with prepositional datives (3).

- (2) a. *He donated me it. *recipient > theme DOC*
 b. *He donated it me. *theme > recipient*
 c. He donated it to me. *prepositional dative*
- (3) a. He denied me it. *recipient > theme DOC*
 b. He denied it me. *theme > recipient*
 c. *He denied it to me. *prepositional dative*

The distribution of the *theme > recipient* order DOCs is dependent on the presence of the *recipient > theme* order DOCs, with all speakers of Northern English having both, while English speakers with only the *recipient > theme* order are the norm. This suggests that the *theme > recipient* order is derived from the *recipient > theme* order.

The structural and c-command relations between objects in the *recipient > theme* order are well-established (Barss and Lasnik 1989, Larson 1988), reflected in the low applicative structure in (4) (Pylkkänen 2008) in which the recipient and theme are introduced within a single Applicative (Appl) projection.

- (4) [_{VP} v [_{VP} V [_{ApplP} RECIPIENT_{IO} [_{Appl'} Appl THEME_{DO}]]]

3. Object Size

Recipient > theme DOCs—in Standard US and Northern English—accept DP or pronominal objects in either position. However, a pronominal theme after a DP recipient is typically unacceptable (5).

- (5) *John gave the girl it. *recipient > theme*

Theme > recipient DOCs show stronger restrictions on object size combinations, which differ according to the specific subvariety of English. The four logical combinations of DPs and pronouns (PRO) differ in their distribution and acceptability. The varieties of English and the permissible object size combinations are summarized in Table 1. I have introduced names for the two subvarieties of Northern English: ‘Core’ and ‘Peripheral’, referencing their geographical distribution.

The key observation is that the theme in the *theme > recipient* order must be a pronoun—specifically a weak pronoun: *me, you, him, her, it, us, them*—and DP themes are unacceptable in all cases (Biggs 2018, Haddican and Holmberg 2019).

4. Hierarchy Effects and the Person Case Constraint

Standard US English has been noted as displaying hierarchy effects in *recipient > theme* DOCs (Bonet 1991, Haspelmath 2004, Anagnostopoulou 2008). Northern English shows the presence of hierarchy effects in both DOC orders.

- (9) a. They showed me *you/*it/*her/*him. 1 > 2/3 *theme* > *recipient*
 b. They showed you ?me/*it/*her/*him. 2 > 1/3 *theme* > *recipient*

4.3 Identical PCC Patterns

The hierarchy effects in the two DOC orders are summarized in Table 2, showing which combinations are acceptable. The data for both DOC orders align with the Ultrastrong PCC (Nevins 2007).

IO	DO	Ultrastrong	Theme > Recipient	Ultrastrong	Recipient > Theme
1	2	✓	? <i>He showed you me.</i>	✓	<i>He showed me you.</i>
1	3	✓	<i>He showed her me.</i>	✓	<i>He showed me her.</i>
2	1	-	* <i>He showed me you.</i>	-	* <i>He showed you me.</i>
2	3	✓	<i>He showed her you.</i>	✓	<i>He showed you her.</i>
3	1	-	* <i>He showed me her.</i>	-	* <i>He showed her me.</i>
3	2	-	* <i>He showed you her.</i>	-	* <i>He showed her you.</i>
3	3	✓	<i>He showed it her.</i>	✓	<i>He showed her it.</i>

Table 2: Overview of PCC Patterns in Northern English.

5. Stress Repair

Stress on the second object repairs unacceptability of both object size and PCC violations.

5.1 Object Size Repair

In *recipient > theme* orders the unacceptable DP-PRO combination is improved with stress (10b).

- (10) a. *They showed the girl him *recipient > theme*
 b. They showed the girl *HIM* *recipient > theme*

This aligns with Anttila et al. (2010) that such combinations are unacceptable due to the theme forming an unstressed foot, which the application of stress repairs.

Stress on a pronominal recipient in the *theme > recipient* order does not repair the DOC (11). Instead the stressed example acquires a *recipient > theme* order reading.

- (11) *I have a brother that I wanted the girl to meet, so...*
 a. ...I showed him her. PRO-PRO *theme > recipient*
 b. *...I showed my brother *HER*. DP-PRO *theme > recipient*

5.2 PCC Repair

When the theme in a *recipient > theme* DOC bears stress previously unacceptable person combinations become acceptable (12).

- (12) They gave you *me/ME. *recipient > theme*

Neither stress on the recipient in *recipient > theme* order nor stress on either object in the *theme > recipient* order repairs hierarchy effects (13b).

- (13) Context: *The computer didn't know what a linguist looked like, so...*
- a. *...they showed *IT* me. *recipient > theme*
 - b. ...they showed it *ME*. *recipient > theme*
 - c. *...they showed me *IT*. *theme > recipient*
 - d. *...they showed *ME* it. *theme > recipient*

Object size restriction and PCC violation can be repaired at the same time (14).

- (14) a. *They showed the student me. *recipient > theme*
b. They showed the student *ME*. *recipient > theme*

6. Agree and Movement

Current models of the PCC requires both objects to Agree with a single head (Nevins 2007, Coon and Keine 2021, Deal 2023, Sichel and Toosarvandani 2024). I propose that *theme > recipient* order DOCs result from the reordering of weak pronouns during the instantiation of the PCC. The mechanism found in Agree-based accounts of the PCC removes intervention-based locality and allows the theme to move over the recipient. Evidence that this movement has occurred comes from:

- i. The inability of the recipient to bind the theme in *theme > recipient* DOCs.
- ii. Onward A-movement for theme passivization.
- iii. Onward $\bar{A}$ -Movement for theme-*wh* superiority.

6.1 Binding

Only the binding diagnostic in Barss and Lasnik (1989) can be implemented due to the presence of object size restrictions in *theme > recipient* DOCs. A naive attempt to apply this diagnostic to the *theme > recipient* order yields ambiguous results: (15a) has the same string and meaning as the *recipient > theme* DOC. The theme restriction excludes testing for the recipient binding a theme anaphor (15b).

- (15) a. !She showed me myself. *theme > recipient*
b. *She showed myself me. *theme > recipient*

However, an R-expression recipient cannot be co-referent with the theme (16), showing that the theme c-commands the recipient in *theme > recipient* DOCs.

- (16) a. I showed him_{*i/j} a friend of John_i. *theme > recipient*

- b. I gave it_{i/j} the owner of the wallet_i. *theme > recipient*

6.2 Onward A-movement: Theme Passivization

Standard US English typically only allows passivization of the recipient object in a DOC, while Northern English allows passivization of either object (17).

- (17) a. I_R was given it_{Th}. *recipient passive* ✓ Standard / ✓ Northern
 b. It_{Th} was given me_R. *theme passive* ✗ Standard / ✓ Northern

Theme passivization in Northern English suggests that the theme is higher than the recipient at some point in the derivation of the *theme > recipient* order. From this position it can undergo passivization without the recipient intervening. Not all speakers of Northern English who have *theme > recipient* DOC accept theme passives (Haddican and Holmberg 2012). However, the acceptance of theme passives in English depends on the presence of *theme > recipient* DOCs. Peripheral Northern cannot have a theme passive with a DP recipient: **It was given the girl*.

DP themes are acceptable in theme passives, despite the corresponding *theme > recipient* DOCs being unacceptable (18).

- (18) a. I gave it her. → It was given her. (PRO-PRO)
 b. *I gave the book her. → The book was given her. (DP-PRO)

6.3 Onward $\bar{A}$ -Movement

Wh-movement is relativized to a nominal bearing a [*uwh*] feature. A lower theme object can be moved to form a *wh*-question without the recipient intervening. Standard US English allows *wh*-extraction of either object in a DOC construction (19).

- (19) a. Who did John give the money? *recipient wh question*
 b. What did John give me? *theme wh question*

However, if both objects are *wh*-words, intervention-based locality once again occurs. Only the higher object bearing the [*uwh*] feature can be moved to form a question: *wh*-superiority. Standard US English has only recipient *wh*-superiority, while Northern English also allows for theme *wh*-superiority (20).

- (20) a. Who did you give what? ✓ Standard / ✓ Northern
 b. What did you give who? ✗ Standard / ✓ Northern

Again, this shows that the theme obtains a position higher than the recipient from which *wh*-movement can subsequently occur without intervention.

6.4 Limits of Movement

The derived theme position being inside the ν P is shown by the verb, theme and recipient remaining in the same relative position to other constituents as in *recipient > theme* DOCs. Manner adverbs occur in the same positions in *theme > recipient* DOCs: a manner adverb appears either before or after the verb and both objects (21). A position between the verb and objects is not possible. As manner adverbs are a ν P adjunct nothing can have raised out of ν P.

- (21) a. He [ν P (slowly) [ν showed it_{Th} me_R] (slowly)]. *theme > recipient*
 b. *He showed [ν P (slowly) [ν t it_{Th} me_R]].
 c. *He showed it_{Th} [ν P (slowly) [ν t t me_R]].

The evidence does not definitively locate the moved theme, but it does show that whatever movements occurs is local to the ν P.

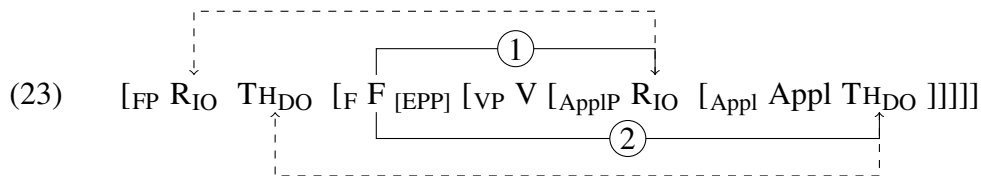
7. Proposal

My proposal implements the Agree-based model of the PCC in Sichel and Toosarvandani (2024). It adopts the ϕ -feature geometry of Harley and Ritter (2002), with the person entailments simplified as the hierarchy in (22).

- (22) *The Person Hierarchy*: [PERSON 3 [PARTICIPANT 2 [SPEAKER 1]]]

In the PCC model of Sichel and Toosarvandani (2024) a ϕ -probe is ‘activated’ through Agree with its first object, and subsequent goals must bear a subset of the first object’s features. Objects are probed in the order IO > DO with the probe on ν , c-commanding both objects. Any object which Agrees can be moved to Spec, ν P, and PCC patterns emerge through the inability of a second object to Agree after probe activation by the first. The Ultrastrong PCC falls out naturally as a consequence of the model as the theme must be equal to or lower on the person hierarchy than the recipient.

For Northern English the probing head cannot be ν given that objects remain within ν P. I instead propose a functional head F which probes, with the potential for movement to Spec, FP. The derivation in (23) shows both the order of probing by F and the potential landing sites for moved objects with tucking in (Richards 1997).

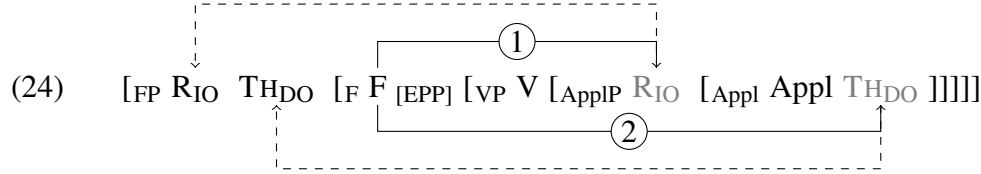


Movement can only occur if motivated and in English DOCs it is caused by either an EPP feature on F or the needs of the theme to move to a structurally-higher position. My analysis focusses on weak pronominal themes, and I propose that weak themes must move

from their base position to avoid stress. Allowing for the EPP feature to be variably present on F provides the key difference between the derivation of *recipient* > *theme* and *theme* > *recipient* order DOCs.

7.1 *Recipient* > *Theme* DOCs

To derive *recipient* > *theme* DOCs Agree must result in movement of both goals where possible. F bears an [EPP] feature and the recipient always move to Spec, F. The theme also moves if it is a weak pronoun to avoid stress (24).



Under this arrangement the different base combinations of objects result in the movement patterns and surface orders shown in Table 3.

	Base Order		Movement		Surface
a.	F [PRO _R PRO _{Th}]	→	PRO _R PRO _{Th} F [__ __]	→	PRO _R PRO _{Th}
b.	F [PRO _R DP _{Th}]	→	PRO _R F [__ DP _{Th}]	→	PRO _R DP _{Th}
c.	F [DP _R DP _{Th}]	→	DP _R F [__ DP _{Th}]	→	DP _R DP _{Th}
d.	F [DP _R PRO _{Th}]	→	DP _R PRO _{Th} F [__ __]	→	DP _R PRO _{Th}

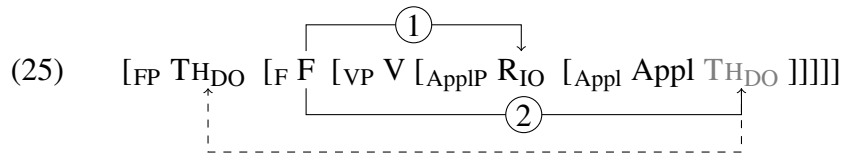
Table 3: Surface Orders with Both Object Movement

The four base object combinations resolve in the following ways:

- **(a.):** The recipient moves due to EPP and the theme moves as it is weak.
- **(b.) and (c.):** The recipient moves due to EPP and the DP theme stays in place.
- **(d.):** The recipient moves due to EPP and the theme moves as it is weak. This combination later fails for PF reasons (Anttila et al. 2010).

7.2 *Theme* > *Recipient* DOCs

To derive *theme* > *recipient* DOCs Agree only results in the movement of an object if that movement is internally-motivated. F lacks an [EPP] feature and recipients never move. The theme moves if it is a weak pronoun to avoid stress (25).



Under this arrangement the different base combinations of objects result in the movement patterns and surface orders shown in Table 4.

	Base Order	Movement	Surface
e.	F [PRO _R PRO _{Th}]	→ PRO _{Th} F [PRO _R ___]	→ PRO _{Th} PRO _R
f.	F [PRO _R DP _{Th}]	→ F [PRO _R DP _{Th}]	→ PRO _R DP _{Th}
g.	F [DP _R DP _{Th}]	→ F [DP _R DP _{Th}]	→ DP _R DP _{Th}
h.	F [DP _R PRO _{Th}]	→ PRO _{Th} F [DP _R ___]	→ PRO _{Th} DP _R

Table 4: Surface Orders with Theme–Only Movement

The four base object combinations resolve in the following ways:

- **(e.) and (h.):** The theme moves as it is a weak pronoun, raising above the recipient and establishing a *theme > recipient* order.
- **(f.) and (g.):** Neither object moves. The *recipient > theme* order is maintained.

Combination (h.) involves the movement of a weak pronoun over a DP and is only acceptable in Core Northern. Given that movement of the weak pronoun entails Agree, the difference is likely to reside in how hierarchy effects are established.

7.3 Enforcing & Repairing Hierarchy Effects

Enforcement of the observed Ultrastrong PCC pattern in these derivations requires that the theme does not Agree if it is more highly–specified for person than the recipient. Only the PRO–PRO combinations in (a.) and (e.) can exhibit the PCC. In these cases a weak theme cannot move if it fails to Agree and remains in its base position where it cannot receive a valid footing without stress.

The asymmetry of stress repair is also explained as *theme > recipient* orders require the theme to move, and it must first successfully Agree—and therefore not violate the Ultrastrong PCC pattern—and be internally–motivated to move by being weak.

8. Outstanding Questions

1. The theme is motivated to move to avoid weakness, but *wh* and passive movement also require motivation. Nominal licensing is likely implicated in theme movement.
2. The variable presence of theme passives among speakers with *theme > recipient* DOCs requires principled explanation.
3. The reason for the different recipient size acceptability in Northern English subvarieties could be due to probe insensitivity to distinctions between third person and δ (Sichel and Toosarvandani 2024), allowing for movement of a pronoun over a DP.

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